Algebra SEP Worksheet 9

8/16/22

Miscellaneous/Review

Arcana

∗ **1.**

On this problem, only the answer will be graded.

(a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?

(b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)

(c) The Jordan form of an operator $A : \mathbb{C}^6 \to \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

∗ **2.**

On this problem, only the answer will be graded. Consider the symmetric group S_4.

(a) What is the order of S_4?

(b) What is the order of its center $Z(S_4)$?

(c) What is the order of its commutator subgroup $[S_4, S_4]$?

(d) Is S_4 a simple group?

(e) Is S_4 a solvable group?

(f) How many conjugacy classes does S_4 have?

∗ **3.**

On this problem, only the answers will be graded.

(a) Give an example of a simple ring which is not a commutative ring.

(b) Give an example a ring R, R-modules M_1, M_2, M_3, and N and a short exact sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ such that the corresponding sequence $0 \leftarrow \mathrm{Hom}(M_1, N) \leftarrow \mathrm{Hom}(M_2, N) \leftarrow \mathrm{Hom}(M_3, N) \leftarrow 0$ is not exact.

Scattered Esoterica

∗ **4.**

Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

(a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$-algebra homomorphism $\phi : A \to B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

(b) Prove that the homomorphism ϕ from part (a) is injective.

(c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

∗ **5.**

Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

(a) Show that R is generated as a $\mathbb{C}$-algebra by $x + y$ and xy.

(b) Show that the map $\mathbb{C}[u, v] \to R$ sending u to $x + y$ and v to xy is an isomorphism.

(c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$-algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

∗ **6.**

Let R be a commutative ring and M an R-module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

(a) Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of R-modules, and let P be an associated prime of M. Prove that P is an associated prime of either M' or M''.

(b) Is the converse true? That is, if P is an associated prime of either M' or M'', must it be the case that P is an associated prime of M?

∗ **7.**

Let L/K be a Galois extension of fields with Galois group G. Let S be the subset of roots of unity in L; that is, S is the set of elements $x \in L$ such that $x^k = 1$ for some positive integer k.

(a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S.)

(b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S.

(c) Give an example of L/K such that the action of G on S is not faithful.

(d) Suppose that the action of G on S is transitive. Prove that $\mathrm{char}(K) = 2$.